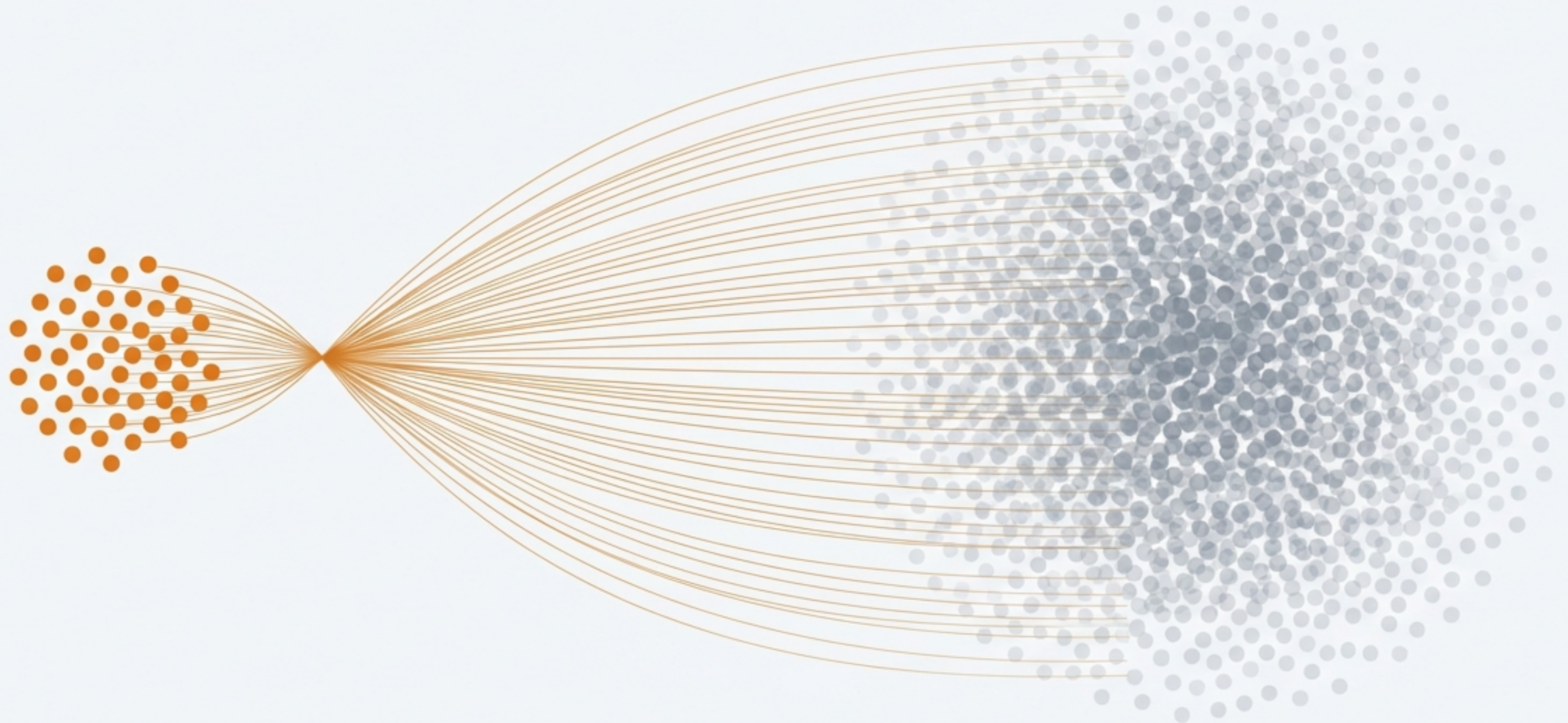


Making Big Conclusions from Small Data

A Practical Guide to Hypothesis Testing for Data Scientists

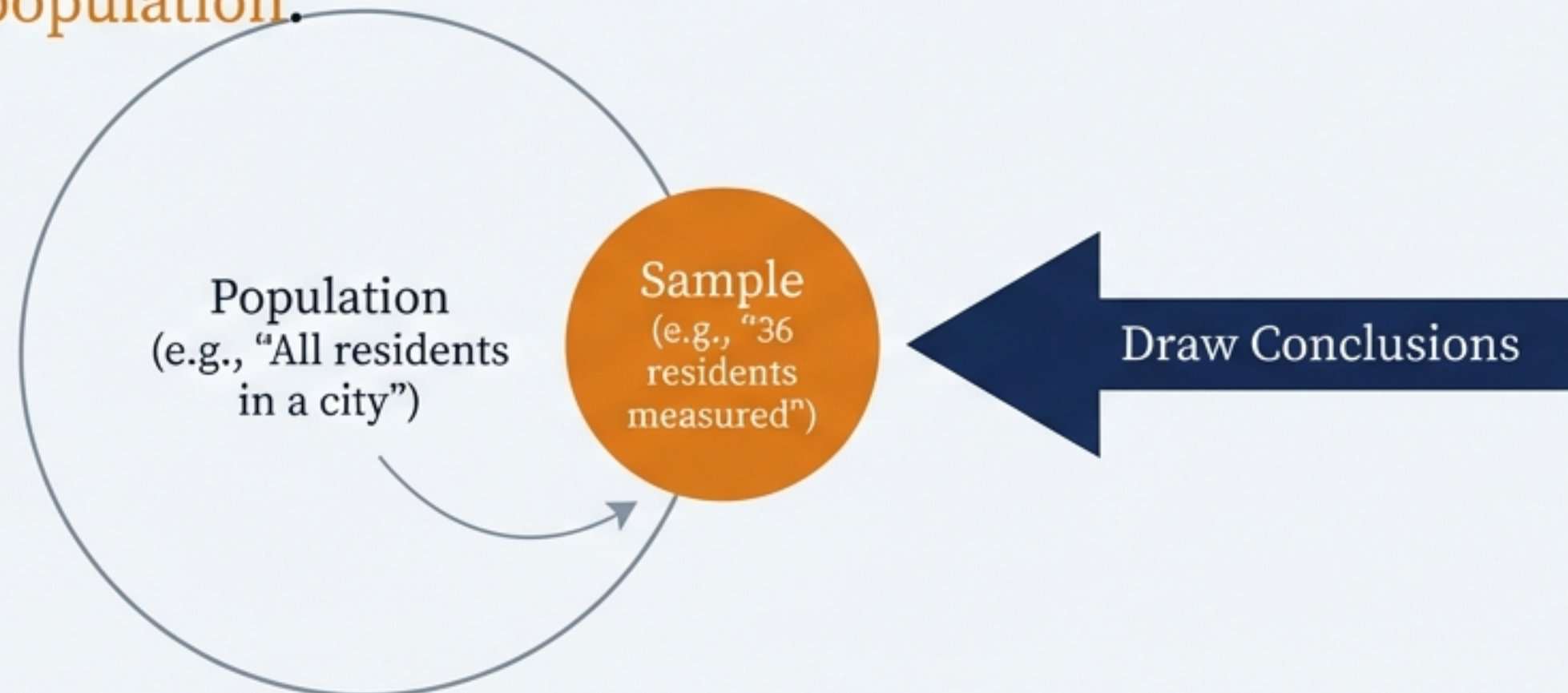


Why We Can't Just Measure Everything

In the real world, we rarely have access to the entire population's data. It's often too expensive, time-consuming, or impossible to collect.

Key Concept: Inferential Statistics

- We take a smaller **sample** of data.
- Through experiments and statistical tests (like Z-Tests and T-Tests), we draw conclusions about the entire **population**.



The Core of the Test: Stating Your Hypotheses

Hypothesis testing is a formal procedure for investigating our ideas about the world using statistics. It always starts with two competing statements.



1. The Null Hypothesis (H_0)

- This is the default assumption, the status quo. It states there is no effect or no difference.
- **Example:** “The average height of all residents in the city **is** 168 cm.”



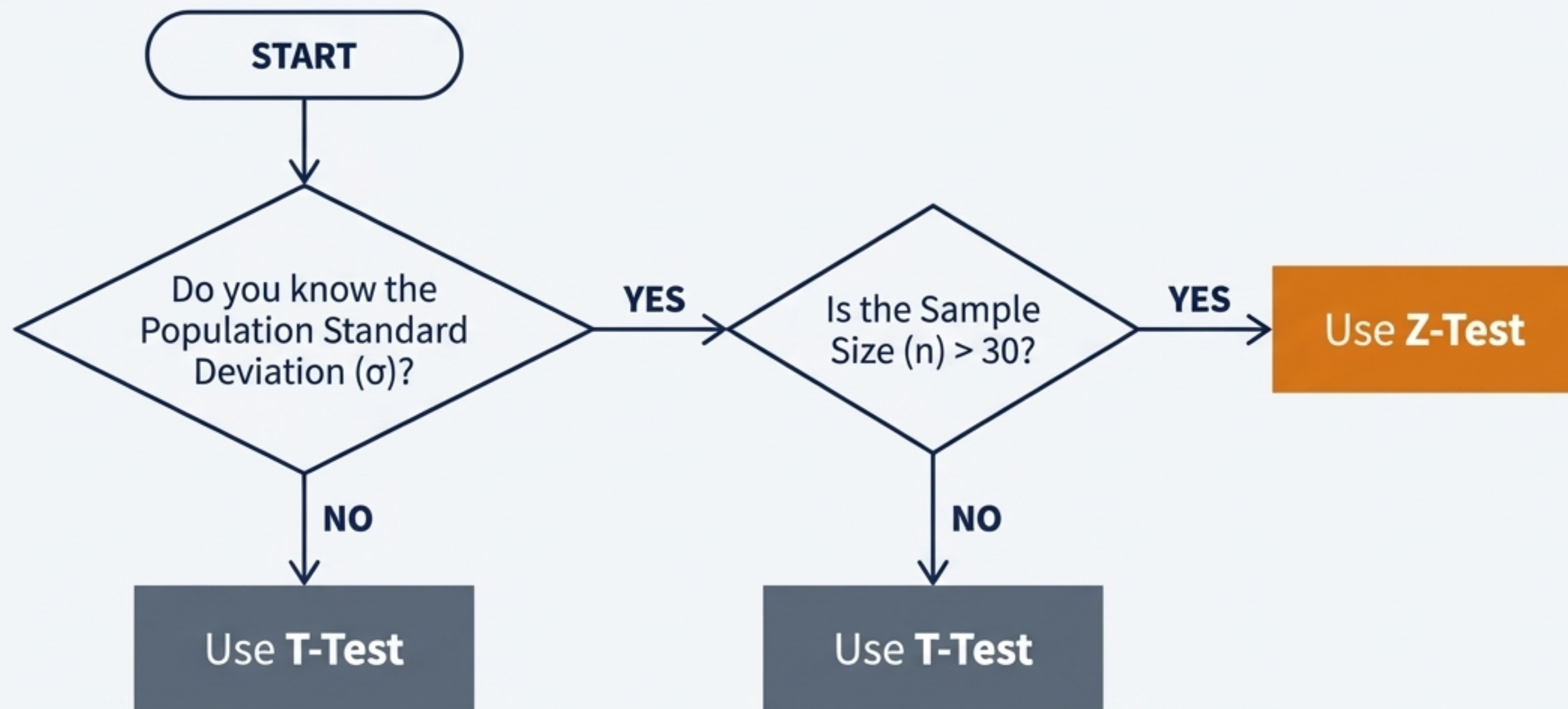
2. The Alternate Hypothesis (H_1)

- This is what you are trying to prove. It's the reverse (or inverse) of the null hypothesis. It challenges the status quo.
- **Example:** “The average height of all residents **is not** 168 cm.”

Key Takeaway: The goal of the test is to see if we have enough evidence in our sample data to reject the null hypothesis.

Choosing Your Tool: When to Use a Z-Test vs. a T-Test

The first step in any analysis is selecting the right statistical tool. This simple flowchart is your guide.



This is one of the most common statistical interview questions. Knowing this decision process is fundamental.

Case Study 1: The Average Height Problem (Z-Test)

The Scenario

The accepted average height of all residents in a city is 168 cm, with a known population standard deviation of 3.9 cm. A doctor believes this average is outdated and different now.

The Experiment

The doctor measures a sample of 36 individuals and finds their average height to be 169.5 cm.

The Question

Is there enough statistical evidence to reject the claim that the city's average height is still 168 cm?

Data Summary

- Population Mean (μ): **168 cm**
- Population Standard Deviation (σ): **3.9 cm**
- Sample Size (n): **36**
- Sample Mean ($\bar{x}$): **169.5 cm**
- Confidence Level: **95%**

Step 1 & 2: Set Up the Hypotheses and Decision Boundary

Step 1: State the Hypotheses

Null Hypothesis (H_0): The population mean height is 168 cm. ($\mu = 168$)

Alternate Hypothesis (H_1): The population mean height is NOT 168 cm. ($\mu \neq 168$)

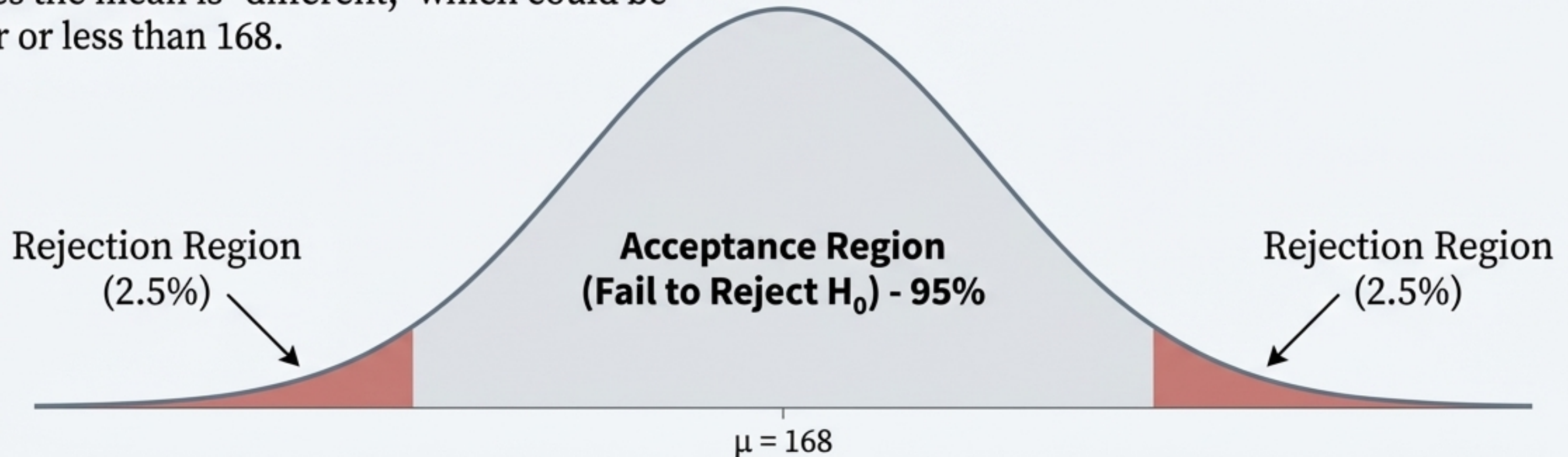
This is a **Two-Tailed Test** because the doctor believes the mean is “different,” which could be greater or less than 168.

Step 2: Define the Confidence Interval & Significance

We want to be **95% confident** in our conclusion.

Confidence Interval (CI) = 95% (0.95)

Significance Level (α) = $1 - \text{CI} = 5\%$ (0.05). This 5% is the total area of our “rejection region.”



Step 3: Find the Critical Value with a Z-Table

Goal

We need to find the Z-scores that mark the edges of our 95% acceptance region.

The Logic

1. Since it's a two-tailed test, our 5% significance is split: 2.5% in the left tail and 2.5% in the right tail.
2. To find the upper critical value, we look for the area to the left of it, which is the entire curve (100%) minus the upper tail (2.5%), or 97.5% (0.9750).
3. We look up the area **0.9750** inside a Z-table.

The Result

The Z-table shows that an non prambless:

- The Z-table shows that an area of 0.9750 corresponds to a Z-score of **1.96**.
- Because the distribution is symmetrical, the lower critical value is **-1.96**.

Conclusion

Our decision boundary is **[-1.96, +1.96]**. If our If our calculated Z-score falls within this range, we fail to reject the null hypothesis. If it falls outside, we **reject it**.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.5	0.0000	0.0000	0.0400	0.9450	0.9450	0.9450	0.9550	0.9550	0.9550	0.9600
1.6	0.9610	0.9630	0.9630	0.9650	0.9600	0.9600	0.9630	0.9600	0.9600	0.9650
1.7	0.9650	0.9700	0.9700	0.9700	0.9700	0.9730	0.9720	0.9720	0.9730	0.9730
1.8	0.9700	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750
1.9	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750	0.9750
2.0	0.9810	0.9820	0.9820	0.9810	0.9820	0.9830	0.9820	0.9830	0.9830	0.9820
2.0	0.9820	0.9830	0.9830	0.9830	0.9835	0.9835	0.9835	0.9835	0.9835	0.9835
2.1	0.9930	0.9930	0.9930	0.9940	0.9945	0.9945	0.9975	0.9975	0.9950	0.9945
2.2	0.9920	0.9930	0.9940	0.9945	0.9940	0.9950	0.9950	0.9940	0.9945	0.9950
2.2	0.9000	0.9000	0.9000	0.9950	0.9950	0.9950	0.9970	0.9950	0.9960	0.9970

$$1.9 + 0.06 = 1.96$$

Step 4 & 5: Calculate the Z-Score and Make a Conclusion



Step 4: Calculate the Test Statistic (Z-Score)

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- $Z = 169.5 - 168 / (3.9 / \sqrt{36})$
- $Z = 1.5 / (3.9 / 6)$
- $Z = 1.5 / 0.65$
- **Z-score = 2.31**

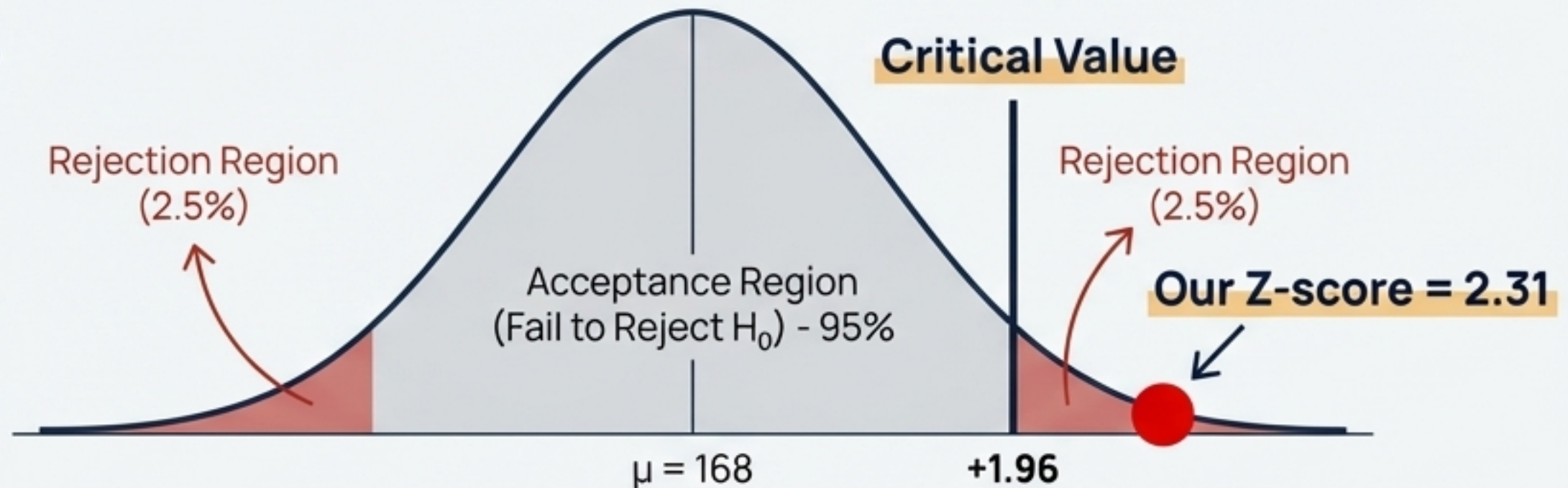


Step 5: Compare and Conclude

Our calculated Z-score is **2.31**.

Our critical value boundary is **±1.96**.

Since **2.31 > 1.96**, our result falls into the rejection region.



The Verdict: We **reject the null hypothesis**. There is enough statistical evidence to support the doctor's belief that the average height in the city is no longer 168 cm.

Case Study 2: The IQ Medication Trial (T-Test)



The Scenario

The average IQ in the general population is 100. A team of researchers develops a new medication they believe could have a positive or negative effect on intelligence.

The Experiment

They give the medication to a sample of 30 participants. After the trial, the sample's mean IQ is 140, with a sample standard deviation of 20.

The Question

Did the medication have a statistically significant effect on intelligence?

Data Summary

- Hypothesized Population Mean (μ): **100**
- Sample Standard Deviation (s): **20** *Note: Population σ is unknown*
- Sample Size (n): **30**
- Sample Mean ($\bar{x}$): **140**
- Confidence Level: **95%**

Solving the IQ Problem: The T-Test in Action



1. Hypotheses:

- H_0 : The mean IQ is **100** ($\mu = 100$).
- H_1 : The mean IQ is **not 100** ($\mu \neq 100$).



2. A New Concept: Degrees of Freedom (df):

- For a one-sample t-test, the formula is simple: **$df = n - 1$** .
- In our case: $df = 30 - 1 = \mathbf{29}$.
- We use df and our significance level ($\alpha = 0.05$) to find the critical value from a t-table. The critical t-value is **± 2.045** .



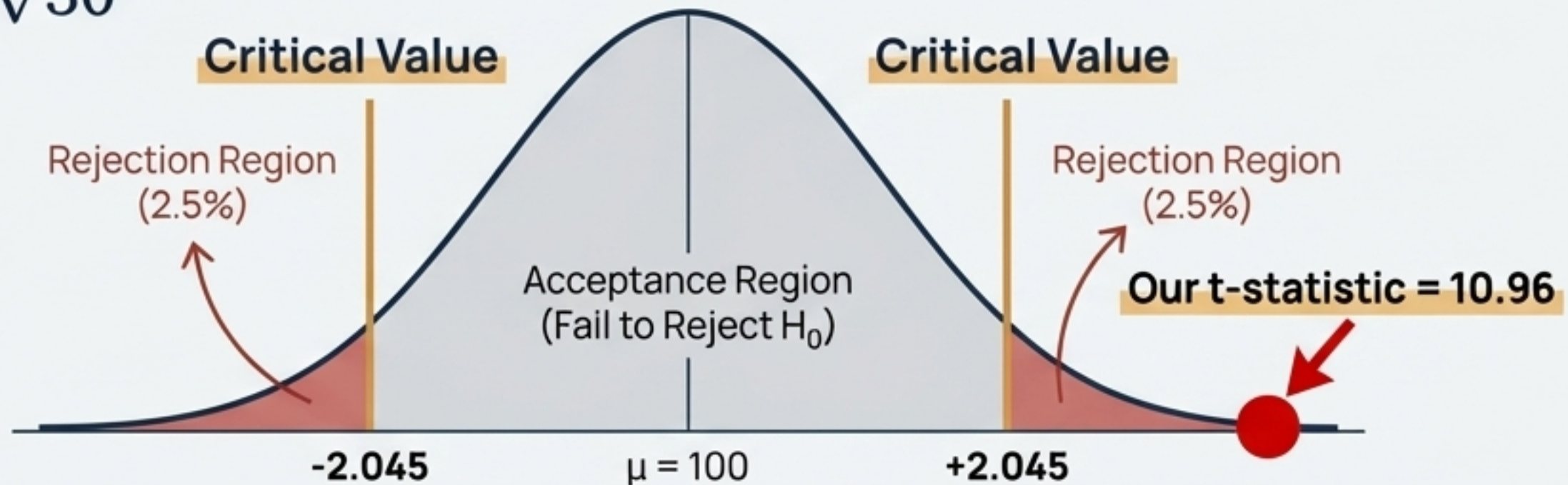
3. Calculate the T-Statistic:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{140 - 100}{20/\sqrt{30}} = 40 / (1.130) = 40 / 3.65 = \mathbf{10.96}$$



4. Conclusion:

- Our calculated t-statistic is **10.96**.
- Our critical value is **± 2.045** .
- Since **$10.96 > 2.045$** , the result is far into the rejection region.



The Verdict: We **reject the null hypothesis**. The medication had a statistically significant effect on IQ scores.

The Risks of Being Wrong: Type I and Type II Errors

No statistical test is perfect. There's always a chance our conclusion is incorrect. Understanding the two types of errors is key to interpreting results responsibly.

	The Reality: H_0 is True	The Reality: H_0 is False
Your Decision: Fail to Reject H_0	 Correct Decision (True Negative) Reality: H_0 is true. Decision: You correctly didn't reject it.	 Type II Error (False Negative) Reality: H_0 is false. Decision: You failed to reject it. You missed a real effect.
Your Decision: Reject H_0	 Type I Error (False Positive) Reality: H_0 is true. Decision: You incorrectly rejected it. You claimed an effect that isn't real.	 Correct Decision (True Positive) Reality: H_0 is false. Decision: You correctly rejected it.

Beyond the Test: Estimating the True Value with Confidence Intervals

What is a Confidence Interval (CI)?

- A Confidence Interval gives a range of values that is likely to contain the true population parameter (e.g., the true population mean).
- It provides an 'interval estimate' rather than a single 'point estimate' (like the sample mean).

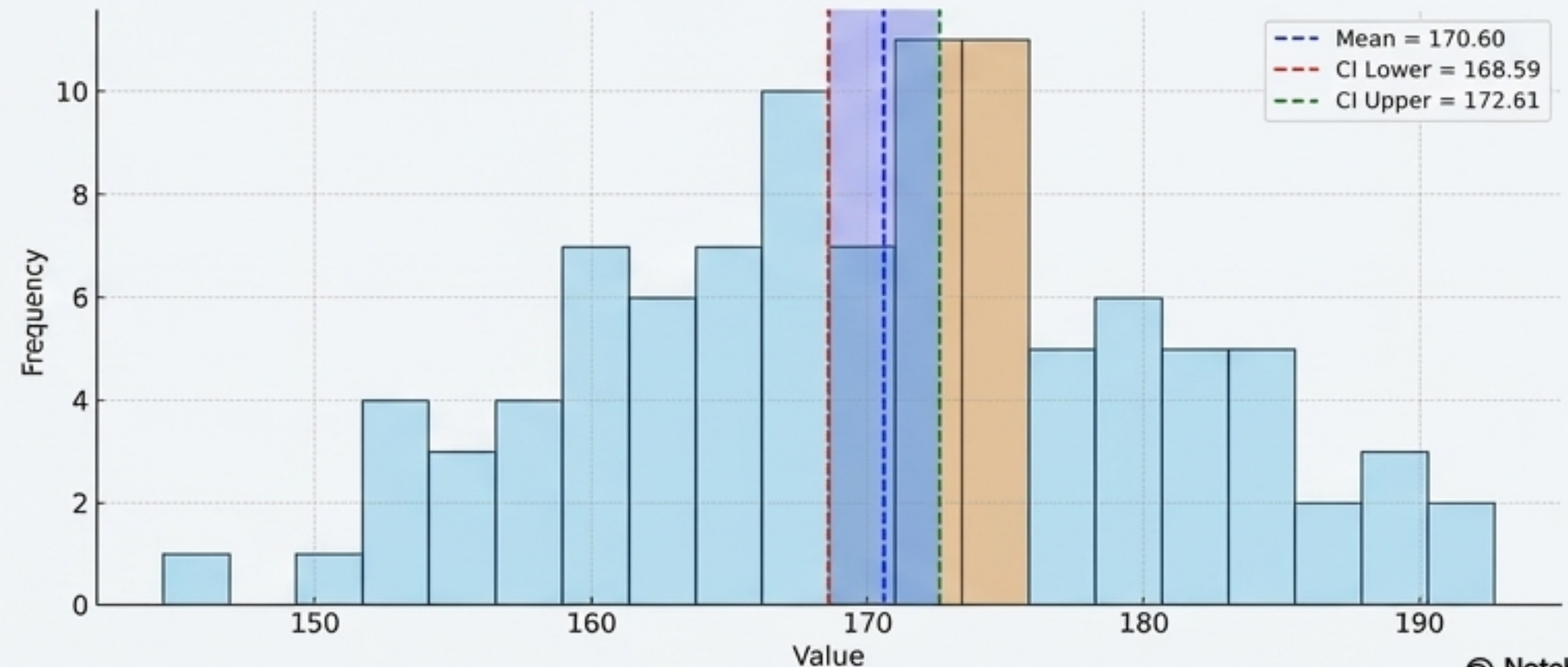
How to Interpret It

- A 95% Confidence Interval means we are 95% confident that the true population mean falls within our calculated range.
- Example: 'We are **95% confident** the true average height for the **entire city** is between 165 cm and 175 cm.'

$$\text{Confidence Interval} = \text{Point Estimate} \pm \text{Margin of Error}$$

Our sample statistic
(e.g., sample mean, $\bar{x}$).

How much we expect our sample result to vary from the true population value. It's calculated using our critical value and standard error.



Calculating a Confidence Interval: A CAT Score Example

Problem: On the verbal section of the CAT exam, the population standard deviation is known to be 100. A sample of 25 test-takers has a mean score of 520. Construct a 95% confidence interval for the true mean score.

1: Identify Components

- Sample Mean ($\bar{x}$) = **520**
- Population SD (σ) = **100**
- Sample Size (n) = **25**
- Z-critical value for 95% CI ($Z_{\alpha/2}$) = **1.96**

2: Calculate Margin of Error

- Margin of Error = $Z_{\alpha/2} * \frac{\sigma}{\sqrt{n}}$
- Margin of Error = $1.96 * \frac{100}{\sqrt{25}}$
- Margin of Error = $1.96 * 20 = \mathbf{39.2}$

3: Construct the Interval

- Lower Bound = $\bar{x} - \text{Margin of Error} = 520 - 39.2 = \mathbf{480.8}$
- Upper Bound = $\bar{x} + \text{Margin of Error} = 520 + 39.2 = \mathbf{559.2}$

We are 95% confident that the **true population mean score** on the CAT verbal section is between **480.8 and 559.2**.

Your Hypothesis Testing Toolkit: A Quick Reference

Which Test to Use?

Z-Test

Use when Population Standard Deviation (σ) is **known** and sample size (**n**) is ideally ≥ 30 .

T-Test

Use when Population Standard Deviation (σ) is **unknown** (you use sample SD "**s**" instead) or when **n** < 30.

Key Terminology Recap

- **Null Hypothesis (H_0)**: The default assumption of no effect.
- **Alternate Hypothesis (H_1)**: The claim you want to test.
- **Significance Level (α)**: The probability of making a Type I error (usually 0.05).
- **Confidence Interval**: The range you believe contains the true population parameter.
- **Type I Error: False Positive**.
Rejecting a true null hypothesis.
- **Type II Error: False Negative**.
Failing to reject a false null hypothesis.

Test Formulas

Z-Score

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

T-Statistic

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Put Your Knowledge to the Test



Your Turn: Use the concepts from this guide to solve the following problem.

The Problem

- A factory manufactures light bulbs with an average warranty of **5 years** and a population standard deviation of **0.5 years**.
- A worker believes a new manufacturing process has reduced the lifespan.
- They test a sample of **40 bulbs** and find the average warranty to be **4.8 years**.

Your Task

- ☐ 1. State the null and alternate hypotheses. (Hint: The worker believes the lifespan is *less than* 5 years, so this is a one-tailed test).
- ☐ 2. Is this a Z-Test or a T-Test?
- ☐ 3. Using a **2% significance level ($\alpha = 0.02$)**, determine if there is enough evidence to support the worker's belief.